

# CS360 Introduction to Database

## Description

Database systems serve as essential back-end infrastructure across a wide range of industries, including finance, e-commerce, retail, manufacturing, and healthcare. They also continue to play a critical role in modern AI applications. This course aims to provide a foundational understanding of what a database is, what a database management system (DBMS) is, what database queries are, and how a DBMS processes queries. In particular, the course focuses on the relational data model, relational databases, relational database management systems (RDBMS), and relational queries (i.e., SQL).

The course is structured into two parts. Part 1 introduces the fundamental concepts of databases and queries, while Part 2 explores how a DBMS processes and executes queries. In addition, students will conduct a term project on Retrieval-Augmented Generation (RAG), a topic that has been gaining increasing importance in industrial applications.

## Lecture Schedule

Part 1:

1. Relational model (Week 1)
2. Design theory for relational databases (Week 2-3)
3. High-level database models (Week 4)
4. Algebraic and logical query languages (Week 5)
5. Database language SQL (Week 6-7)

Part 2:

6. Secondary storage management (Week 9)
7. Index structures (Week 10-11)
8. Query execution (Week 12-13)
9. Query optimization (Week 14-15)

## Textbook

Hector Garcia-Molina, Jeffrey D. Ullman, and Jennifer Widom, *Database Systems – The complete book (2nd ed.)*, Pearson Education Asia, 2011.

## Evaluation Criteria

- Midterm exam: 20%
- Final exam: 25%
- SQL programming assignment: 10%
- DBMS programming assignment: 20%
- RAG programming project: 25% (RAG: Retrieval-Augmented Generation)

**Reading assignments** (Please read the textbook according to the class progress)

- Week 1: ch. 1, ch. 2
- Week 2–3: ch. 3
- Week 4: ch. 4.1 – 4.6
- Week 5: ch. 5
- Week 6: ch. 6.1 – 6.4
- Week 7: ch. 6.5 – 6.6
- Week 9: ch. 13
- Week 10-11: ch. 14
- Week 12–13: ch. 15
- Week 14–15: ch. 16